1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

> *"Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem"*

- Highly relevant: This is the central skill being assessed.
- Each statement must be evaluated using both numerical reasoning and contextual logic:
  - Statement 1: 3 × 3,200,000 = 9,600,000 > 8,000,000 → Yes (direct calculation; valid).
  - Statement 2: Maintenance cost is 0.01 zeds/kWh, revenue is 0.10 zeds/kWh → 0.01 / 0.10 = 10%, not 5% → No (a common trap for inattentive students).
  - Statement 3: Since maintenance cost is expressed as "zeds per kWh," it clearly scales with energy output → Yes.
  - Statement 4 (critical): "97% operational availability" does not mean "97 days offline." With 365 days/year, 3% downtime ≈ 10.95 days—not 97. Mistaking "97%" for "97 days" is a classic conceptual/unit confusion. Correctly answering No demonstrates precise understanding of percentage meaning.
- Especially for Statement 4, the task tests whether students can reject a superficially plausible but fundamentally flawed interpretation, perfectly aligning with Competency #2.

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2. Competency #3: Identify the limits of the model used to solve a problem

> *"Identify the limits of the model used to solve a problem"*

- Highly relevant: The problem presents a set of simplified, static parameters (fixed prices, linear costs), and students must recognize what cannot be assumed:
  - All inferences must rely only on the given data—no external factors (e.g., inflation, equipment degradation) may be introduced;
  - "97% operational time" is a statistical average, not a fixed calendar count; it certainly does not imply exactly 97 non-operational days;
  - Accepting Statement 4 would reflect a failure to respect the boundaries of the model's representation.
- Rejecting Statement 4 correctly shows awareness that "97%" is a proportion of time, not a literal day count—a clear demonstration of Competency #3.

3. Competency #9: Understand the relationship between the problem context and the mathematical representation, and use this to interpret the real-world meaning, feasibility, and limitations of the solution

> *"Understand the relationship between the context of the problem and representation of the mathematical solution... gauge the feasibility and possible limitations of the solution"*

- Strongly relevant: Students must accurately map mathematical expressions to real-world engineering and economic meanings:
  - "0.01 zeds/kWh" explicitly denotes a variable cost, justifying Statement 3;
  - "97% of the year" is a time-based availability metric, requiring conversion via 3% × 365 ≈ 11 days—not misreading "97" as days;
  - The constant ratio of maintenance to revenue (10%) means there is no basis for "approximately 5%."
- This precise translation from symbolic form to contextual meaning is the essence of Competency #9.

4. Competency #1: Interpret a mathematical result back into the real-world context

> *"Interpret a mathematical result back into the real world context"*

- Moderately relevant: For example:
  - Calculating 3 × 3.2M = 9.6M must be understood as "this indeed exceeds 8 million zeds";
  - Recognizing that a 10% maintenance share means "for every 1 zed earned, 0.10 zed is spent on maintenance."
- Though calculations are simple, assigning real-world significance to numbers is essential for accurate judgment.

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5. Competency #8: Recognise and explain the scope and limitations of mathematical concepts and solutions

> *"Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions"*

- Moderately relevant: Particularly evident in interpreting "97%":
  - A percentage is a relative measure, not an absolute count;
  - Students must understand that "97% availability" describes a performance indicator, not a calendar schedule.
- Correctly labeling Statement 4 as No demonstrates awareness of the conceptual boundaries of percentages.

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6. Competency #6: Use real-world knowledge to judge how mathematical results should be adjusted or applied

> *"Use knowledge of how the real world impacts the outcomes... to make contextual judgments"*

- Weakly relevant: All judgments can be made using only the provided data—no external domain knowledge (e.g., actual wind farm operations) is required.
- However, students familiar with energy infrastructure might more confidently reject Statement 4, though such knowledge is not necessary.

1. 第2项：判断数学结果或结论在给定问题情境下是否合理

> *"Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem"*

- 高度相关：这是本题最核心的能力。
- 每个陈述都需要结合数据含义与现实逻辑判断其是否“可推断”：
  - 陈述1：3 × 3,200,000 = 9,600,000 > 8,000,000 → Yes（直接计算，合理）；
  - 陈述2：维护成本 0.01 / 收入 0.10 = 10%，而非 5% → No（常见误判点，需警惕表面近似）；
  - 陈述3：因维护成本以 “zeds per kWh” 给出，显然与发电量成正比 → Yes；
  - 陈述4（关键）：“97% 的年运行时间” ≠ “97 天不运行”。一年 365 天，3% 不运行 ≈ 10.95 天，而非 97 天。将“97%”误解为“97天”是典型单位/概念混淆。正确判断“No”体现对百分比含义的准确理解。
- 尤其是陈述4，直接测试学生是否能识别看似合理但实际荒谬的错误解释，完美契合第2项。

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2. 第3项：识别所用模型的局限性

> *"Identify the limits of the model used to solve a problem"*

- 高度相关：题目提供的是一组简化、静态参数（如固定电价、线性成本），但学生需意识到：
  - 所有计算均基于“给定数据”，不能引入外部假设（如电价波动、设备老化）；
  - “97% 运行时间”是一个统计平均值，不代表每年精确停机 10.95 天，更不等于 97 天；
  - 若学生试图用“可能某年停机97天”来支持陈述4，就忽略了模型的确定性表述边界。
- 正确拒绝陈述4，即表明理解“97% 是比例，不是天数”——这正是识别模型表征局限性的体现。

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3. 第9项：理解问题情境与数学解表示之间的关系，并据此解释解的现实含义及可行性/局限性

> *"Understand the relationship between the context of the problem and representation of the mathematical solution... gauge the feasibility and possible limitations of the solution"*

- 强相关：学生必须将数学表达（如“0.01 zeds/kWh”、“97%”）准确映射到工程经济语境：
  - “per kWh” 明确表示可变成本，故陈述3可推断；
  - “97% of the year” 是时间比例，需转换为天数时应计算 3% × 365 ≈ 11 天，而非误读为 97 天；
  - 收入与成本均为线性函数，因此比率恒定（10%），不存在“约5%”的空间。
- 这种符号→语义的精准转换是第9项的核心。

4. 第1项：将数学结果还原到现实世界的情境中

> *"Interpret a mathematical result back into the real world context"*

- 中等相关：例如，计算 3 × 3.2M = 9.6M 后，需理解“这确实超过 8M”；
  或理解“10% 的维护占比”意味着每赚 1 zed，花 0.1 zed 维护。
- 虽然计算简单，但赋予数字现实意义是判断的基础。

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5. 第8项：认识并说明数学概念及其解的适用范围与局限

> *"Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions"*

- 中等相关：尤其体现在对“97%”的理解上：
  - 百分比是相对量，不能直接等同于绝对天数；
  - 学生需明白“97% operational”描述的是可用性指标，而非日历日计数。
- 正确判断陈述4为“No”即展示了对百分比概念适用边界的认识。

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6. 第6项：利用现实世界知识判断如何调整或应用数学结果

> *"Use knowledge of how the real world impacts the outcomes... to make contextual judgments"*

- 弱相关：本题所有判断均可仅基于给定数据完成，无需外部知识（如风力发电实际运维经验）；
- 但若学生知道“风力电站不可能停机97天/年而仍称97%可用”，会更坚定判断，不过非必需。